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APPROVED METHOD: LIFE TESTING OF FLUORESCENT LAMPS

AN AMERICAN NATIONAL STANDARD

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Prepared By
The IES Testing Procedures Committee

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Foreword

This Approved Method is a revision of IESNA LM-40-2001, *IESNA Approved Method for Life Testing of Fluorescent Lamps*; changes have been made to update information, to give clearer guidelines for test requirements, and to promote uniformity in measurement procedure.

1.0 Introduction and Scope

1.1 Introduction

This guide describes the procedures by which fluorescent lamps can be operated under controlled conditions to obtain optimally comparable data on individual lamp life, changes in light output, and other parameters that vary during the life of the lamp.

The lamp is usually in the form of a long glass tube with an electrode sealed into each end. The inner wall of the tube is coated with one or more fluorescent powders called *phosphors*. The lamp is dosed with a few milligrams of liquid mercury and filled with an inert gas such as argon to a few Torr.* When the proper starting voltage is applied, the inert gas breaks down to form a low current discharge. This discharge increases the temperature of the liquid mercury dose, which vaporizes, and the discharge transforms from a low pressure (2 to 3 Torr), low current, inert gas discharge to a higher pressure (approximately 760 Torr), higher current, mercury discharge. Some visible radiation is produced by emission from the inert gas and the mercury vapor. Some of the mercury in the discharge is excited to a higher-energy state from which it emits UV radiation to excite the phosphors. The phosphors absorb and convert UV radiation to visible radiation, or light. Most of the light is generated from the conversion of the mercury UV emission by the phosphors to visible radiation.

The phosphors in general use have been selected and blended to respond most efficiently to ultraviolet energy at a wavelength of 253.7 nanometers, the principal wavelength generated in a low pressure mercury discharge. Like most electric discharge lamps, fluorescent lamps must be operated in series with a current-limiting device, commonly called a *ballast*. The ballast limits the arc current to the value for which each lamp type is designed. It also provides the required starting and operating lamp voltages.

It is often important to know the light output, efficacy, and lamp lumen maintenance as well as the life of these lamps. The annex to this document discusses this issue. (For information on the photometry of fluorescent lamps see LM-9.[1])

Performance of fluorescent lamps is typically affected by variables such as operating cycle, conditions imposed by the auxiliary equipment and fixture, ambient temperature, vibration, drafts, and orientation. Test conditions and programs should be designed to give comparable results when adopted by various laboratories. The recommendations of this IES approved method have been made with these objectives in mind.

1.2 Scope

This approved method describes the procedures to be followed and the precautions to be observed in obtaining uniform and reproducible measurements during life testing of fluorescent lamps under standard conditions. It addresses life testing of linear, U shaped, and circular shaped fluorescent lamps operated on auxiliary devices designed and certified to meet lamp industry standards and tolerance. This Approved Method does not include single ended compact fluorescent lamps.[2]

2.0 Normative References

Current versions of:

American National Standards Institute. *Line Frequency Fluorescent Lamp Ballasts,* C82.1. New York.

American National Standards Institute. *Electric Lamps - Specifications for Fluorescent Lamp Starters,* C78.180. New York.

* The Torr is a unit of pressure, equal to 1/760 atmosphere.

3.0 Nomenclature and Definitions

The units of electrical measurement used in this approved method are the volt, the ampere, and the watt. Additional definitions may be found in *ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering*.[3]

4.0 Ambient and Physical Conditions

4.1 General

It is good laboratory practice that the storage and testing of lamps should be undertaken in a relatively clean environment.

4.2 Vibration

Lamps should not be subjected to excessive vibration or shock during life testing.

4.3 Temperature

Ambient temperature shall be controlled between 15 ˚C (60 ˚F) and 40 ˚C (105 ˚F). When the testing temperature range is exceeded, life testing shall be suspended.

Temperatures above 40 ˚C (105 ˚F) may be deleterious to ballasts and components and may affect lamp starting. The ballast should be operated at a temperature less than or equal to the maximum recommended case temperature.[4] When specified by the manufacturer, critical lamp temperatures, such as the bulb wall temperature or the control point temperature, shall be maintained.

4.4 Air Movement

Airflow should be minimized for proper lamp starting and operation.

Airflow around the lamps should be such that normal convective airflow is not affected. Because some air movement is necessary to maintain the specified ambient temperature, care should be taken to minimize any drafts directly on the lamps.

4.5 Operating Orientation and Lamp Spacing

The manufacturer should specify the operating orientation of the lamps under test. Where an orientation is not specified, or where more than one orientation is specified, the lamp should be tested in the orientation that will be used in the application.

The lamps shall be spaced to allow airflow around each lamp. This is facilitated by designing open life-testing racks with minimal structural components to block airflow.

5.0 Electrical Conditions

5.1 Input Voltage

Input voltage shall conform to the rated input voltage (root mean square [rms]) and frequency of the auxiliary equipment. If the rated input voltage of the auxiliary equipment is a range, the center value or the voltage specifically requested for the testing shall be used. In all cases, the voltage used shall be reported as a test condition.

5.2 Line Voltage Waveshape

The power supply shall have a voltage waveshape such that the total harmonic distortion does not exceed 3 percent of the fundamental.

5.3 Voltage Regulation

The voltage shall be monitored and regulated to within 2 percent of the rated rms value for the duration of the life test.

5.4 Auxiliary Equipment

5.4.1 Ballast. The type of ballast chosen may affect lamp life and electrical values during life. In general the types of ballasts having the widest use should be chosen for the test. Ballasts used in life tests shall comply with the specifications in ANSI C82.3. If multiple lamp ballasts are used, the number of lamps required by the ballast design shall also be used.

5.4.2 Starters. Starters shall comply with the specifications in ANSI C78.180.

5.5 Instrumentation

Accurate recording of elapsed operating time is important in lamp life testing. If used, an elapsed-time recording device shall accumulate time only when the associated lamps are operating including seasoning. Video monitoring, current monitoring or other means can be used to determine elapsed operating time if designed to provide sufficient temporal accuracy.

To facilitate the periodic checking of auxiliary equipment or lamp characteristics, appropriate test receptacles may be provided to permit instruments to be connected to each circuit. Instruments shall measure true root mean square (rms) values.

5.6 Wiring

All wiring of life test racks should be in accordance with national, state or provincial, and local electrical codes, paying particular attention to the color-coding on the input wiring to the ballast. The ballast shall be grounded per manufacturer's specifications.

5.7 Lamp Connections

When required, the two diagonally opposed contact pins, across which the highest lamp voltage occurs when operating on the life test rack, shall be marked and referred to as the "hot pins." These contact pins shall be connected to any operating circuit so that they remain the hot pins during any testing, such as offline photometric or electrical evaluation performed during life testing. A lamp being returned to a life test position shall have the same two contact pins as hot pins as before the lamp was removed.

An inspection of lamp holder contacts and ballast verification should be performed whenever lamps are placed on the life test rack.

5.8 Adjacent Ground

A starting aid or grounded metal strip (especially for rapid start lamps), at least 25 mm (1 in.) wide and extending essentially the full length of the lamp (cathode to cathode), should be installed behind each lamp under test.[5]

6.0 Lamp Test Procedures

6.1 Lamp Selection

The lamps shall be selected to represent the purpose of the test. However, if the test is designed to evaluate specific characteristics, the sample selection needs to take this into consideration. The value of the test will depend upon the method of sampling, size of the sample, conditions of testing, and other factors.

6.2 Lamp Handling

Prior to evaluation, lamps shall be cleaned to eliminate handling marks.

If the lamps are to be removed from the life test racks for evaluation, handling, transporting or storing, maintaining the lamps in the same orientation as during operation on the life test racks can reduce lamp stabilization time for evaluation and for subsequent continuation of life testing. This practice is generally not a requirement but can be effective and is recommended with lamp types that have defined control points or cold chambers.

6.3 Lamp Marking

Individual lamps shall be tracked during life testing. For example, lamps may be identified by markings applied directly to the lamps, by labels attached to the lamps during transport and evaluation, or by labels attached to the life-test rack position occupied by the lamps during the life test. The identification method selected should take into account the effect of exposure to light, UV and IR radiation, and heat.

Possible suitable marking methods or materials include durable bar coding, ceramic ink marking, high temperature markers, or any other method that endures or can be periodically renewed for the length of the life testing.

6.4 Operating Cycle

The standard operating cycle shall be 180 minutes on, 20 minutes off. Life data from different operating cycles cannot be compared without additional information. There is no industry consensus method for accelerated life testing of fluorescent lamps. For recording of operating time, refer to **Section 5.4**.

6.5 Recording Failures

Checking for lamp failures either by direct observation or automatic monitoring shall be done at an interval of no more than 1 percent of rated life. The recorded failure time shall be determined as the midpoint of the last monitoring interval. Each failure shall be investigated to make certain that it is actually a lamp failure and is not caused by improper functioning of the auxiliary equipment or lamp holders. If multiple-lamp ballasts are used, the failed lamp shall be replaced by an operating lamp of the same lamp type.

Lamp failures can be divided roughly into three classes: those caused by manufacturing defects, those caused by normal failures, and lamps showing markedly low light output.

7.0 Test Report

Test reports shall include the hours at which lamps failed for each lamp installed at the start of the test.

The report shall also list all pertinent data concerning conditions of testing, type of equipment, and lamp types being tested. Typical items reported are:

- Number of lamps tested
- Selection criteria
- Description of lamps
- Description of auxiliary equipment
- Operating cycle
- Ambient conditions
- Input voltage to ballast
- Lamp operating orientation
- Operating hours at which lamp failed including seasoning time
- Class of failure of each lamp
- Lamp monitoring interval
- Median life of entire group
- Special test conditions (example: minimum light output of individual lamps before it is considered a lamp failure; any unusual conditions observed)

Annex – Measuring Lamp Lumen Maintenance

It is often desirable to measure the lamp lumen maintenance during the course of a life test. This function of lamp performance is determined from measurement of the photometric characteristics of the lamp during the life test, typically expressed as a percentage of rated output. Photometric measurements are made after at least two intervals, customarily 40 percent and 70 percent of rated life. These values are compared to the initial lamp output measured after the initial seasoning time of the lamp. More intervals may be used to further define the shape of the lamp lumen-maintenance curve.

INFORMATIVE REFERENCE LIST

1 Illuminating Engineering Society. ANSI/IES LM-9-20/R23, Approved Method: Electrical and Photometric Measurement of Fluorescent Lamps. New York: IES; 2020.

2 Illuminating Engineering Society. ANSI/IES LM-65-20/R23, Approved Method: Electrical and Photometric Measurement of Single-Based Fluorescent Lamps. New York: IES; 2020.

3 Illuminating Engineering Society. ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020.

4 UL, LLC. UL 935, Standard for Fluorescent-Lamp Ballasts. Northbrook, IL: UL, LLC; 2018.

5 American National Standards Institute. C78.81-2005, American National Standard for Electrical Lamps — Double-Capped Flourescent Lamps — Dimensional and Electrical Charateristics. New York: ANSI; 2005.

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